

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Koel Thane - 12 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Koel Thane

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

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Course : Computer networks

Course code : CSA0714.

CO2 - AT-1 - ANALYTICAL

1) Given:

Network : 192.168.20.0/24.

The network is divided into:

LAB A $\rightarrow$ /26

LAB B $\rightarrow$ /27

LAB C $\rightarrow$ /28.

Subnet Allocation:

LAB A (/26)

* Subnet mask : 255.255.255.192

* No. of IP addresses :

$$\rightarrow 2^{(32-26)} = \underline{64}$$

$$\rightarrow 64 - 2 = \underline{62}.$$

* Network address : 192.168.20.0

$\rightarrow$ First Host : 192.168.20.1

$\rightarrow$ Last Host : 192.168.20.62.

Broadcast address : 192.168.20.63.

* Subnet mask : 255 . 255 . 255 . 224 .

* NO of IP addresses:

$$\rightarrow 2^{(32 - 27)} = 32$$

$$\rightarrow 32 - 2 = 30$$

→ First host : 192 . 168 . 20 . 65

→ Last Host = 192 . 168 . 20 . 94 .

∴ Broadcast address : 192 . 168 . 20 . 95

LABC (28)

* Subnet mask : 255 . 255 . 255 . 240

* NO of IP addresses

$$\rightarrow 2^{(32 - 28)} = 16$$

$$\rightarrow 16 - 2 = 14$$

* Network address : 192 . 168 . 20 . 96 .

→ First host : 192 . 168 . 20 . 97

→ Last host : 192 . 168 . 20 . 110 .

∴ Broadcast address = 192 . 168 . 20 . 111

~~Calculations:~~

Conclusion:

* The university network $192.168.20.0/24$ is divided into three subnets for A, B and C labs.

* LAB A receives a $/26$ subnet with 62 host addresses.

* LAB B receives a $/27$ subnet with 30 host addresses.

* LAB C receives a $/26$ subnet with 14 host addresses.

3. Given:

* IP address Block : 172.16.0.0 /16

* Departments and subnet masks :

- HR = /25

- Sales = /26

- IT = /27

- Admin = /28

Subnet Allocation:

1) HR Department (/25).

* Subnet mask : 255.255.255.128.

* Total IP address :

$$* 2^{(32-25)} = \underline{128}$$

$$* 128 - 2 = \underline{126}$$

* Network address : 172.16.0.0

→ First Host : 172.16.0.1

→ Last Host : 172.16.0.126.

Broadcast address : 172.16.0.127.

Sales Department (/26):

* Subnet mask : 255.255.255.192.

* Total IP addresses :

$$* 2^{(32-26)} = \underline{64}$$

$$* 64 - 2 = \underline{62}$$

* Network address : 172.16.0.128.

→ First host : 172.16.0.129

→ Last host : 172.16.0.190.

Broadcast Address : 172.16.0.191.

3) IT Department (/27):

* Subnet mask : 255.255.255.224.

* Total IP addresses :

$$* 2^{(32-27)} = \underline{32}$$

$$* 32 - 2 = \underline{30}$$

→ Network address : 172.16.0.192

* First host : 172.16.0.193

* Last host : 172.16.0.222

Broadcast Address : 172.16.0.233.

4) Admin Department (128) :

* Subnet mask : 255.255.255.240

* Total ip address :

$$* 2^{(32-28)} = \underline{16}$$

$$* 16 - 2 = \underline{14}$$

* network Address : 172.16.0.224 .

→ First host : 172.16.0.225 .

→ Last host : 172.16.0.238 .

Broadcast Address : 172.16.0.239 .